

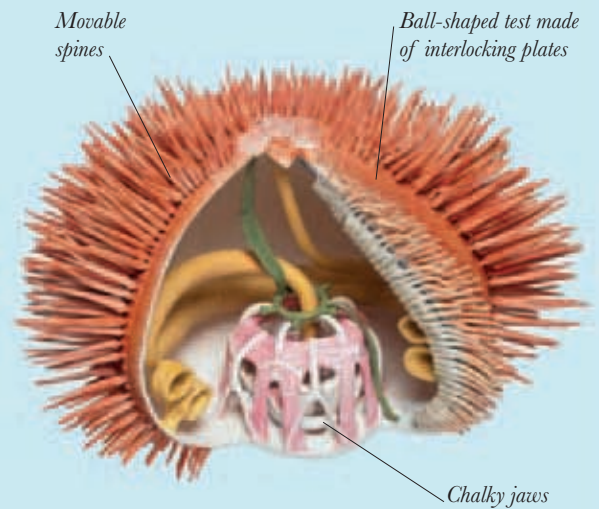


Storm protection

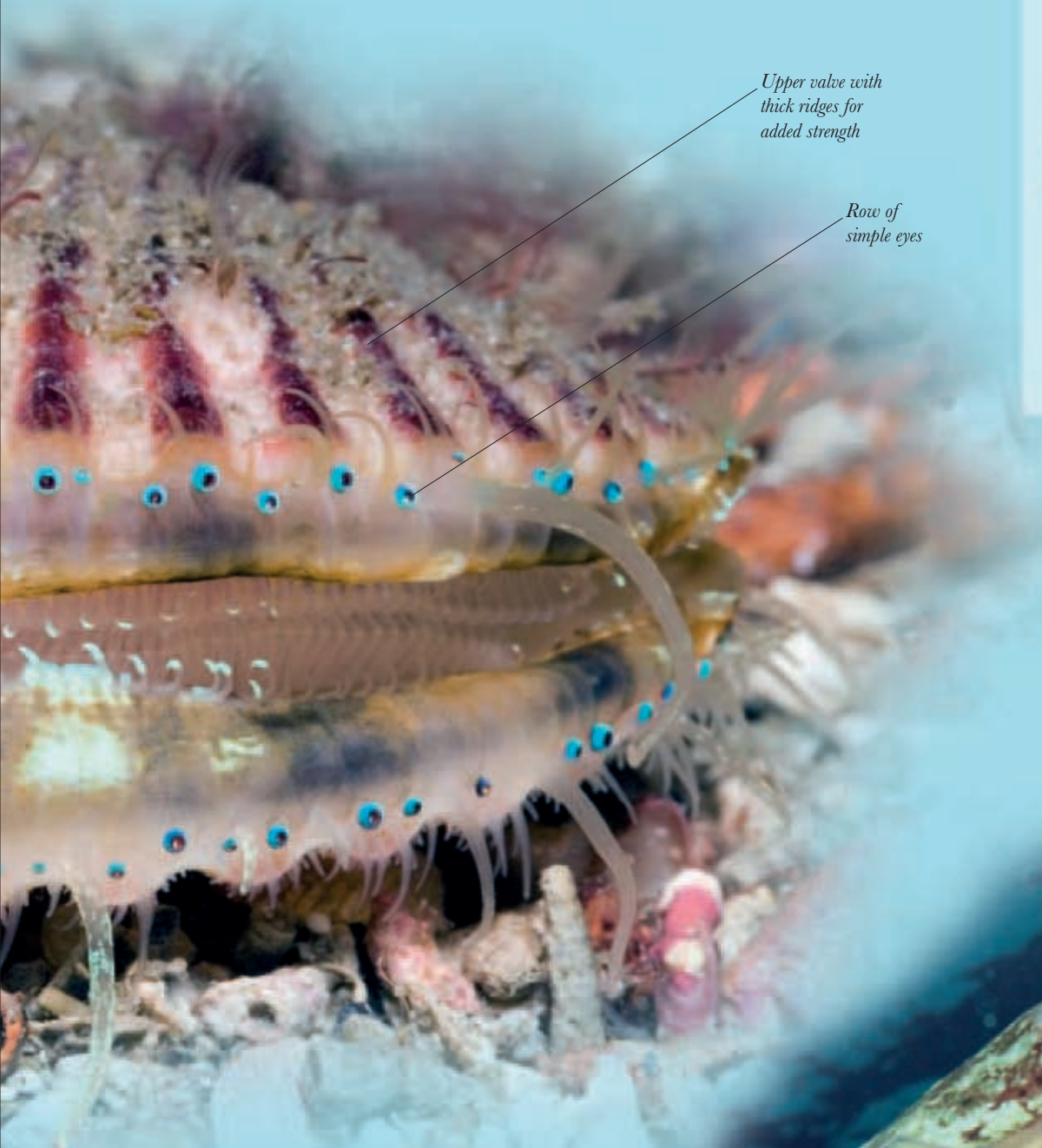
Limpet shells are among the toughest in the animal world. Their conical shape withstands being battered by the waves and also fends off predatory birds. At low tide, these mollusks clamp their shells firmly to the hollows that they grind in rocks. When the tide is high, limpets creep over rocky shores feeding on algae, but they always return to the same spot before the water recedes.

Echinoderm cases

Echinoderms, such as sea urchins and starfish, have small chalky plates set in the outer layer of their skin. A starfish's plates are connected by muscles, which enable its five arms to flex and bend. A sea urchin's plates fit together to form a rigid case called a test, protected by a covering of sharp spines.



Model of a sea urchin test



Upper valve with thick ridges for added strength

Row of simple eyes



Cuttlebone

Inner strength

Unlike the shells of other mollusks, a cuttlefish's shell is hidden inside its body. Known as a cuttlebone, this flat, chalky shell acts as a gas-filled float. A cuttlefish can change the amount of gas in the float, letting it rise through the water when swimming or sink to the seabed. Squid have similar internal shells, known as pens.

